

Week 7 — Demo: Sampling, Spectra, and Isostatic Gravity

Geophysics 3A: Potential Fields & Geothermics

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1 Purpose of the Demonstration

This demonstration extends the basic ideas of Fourier transforms developed in Demo 1 to explore the impacts of sampling on what information can be recovered from series data.

The notebook progresses deliberately from:

- synthetic geological models with known truth,
- explore impact of sampling size,
- see how aliasing can cause misinterpretation.

2 In-Class Demonstration Outline

A suggested walkthrough is:

1. **Synthetic folds:** increase station spacing until aliasing occurs.
2. **Synthetic Basin–Range:** observe dominance of regional signal.
3. **Spectra:** show migration of power with coarser sampling.

3 Sampling and Aliasing in Space

A spatially varying gravity field $g(x)$ can be represented as a sum of spatial frequencies (wavenumbers):

$$g(x) = \int \hat{g}(k) e^{i2\pi kx} dk, \quad (1)$$

where k is the spatial frequency (cycles per unit distance) and the corresponding wavelength is $\lambda = 1/k$.

If the field is sampled at spacing Δx , the Nyquist wavelength is

$$\lambda_N = 2\Delta x. \quad (2)$$

Wavelengths shorter than λ_N cannot be represented correctly. Instead, their energy is mapped to longer wavelengths: this is *aliasing*.

3.1 Key property of aliasing

Aliasing is not noise and not a numerical error. The Fourier transform recovers a *coherent but incorrect* spectrum. Once aliasing occurs, the original short-wavelength information is irretrievably lost.

To put it simply, there are four consequences of aliasing:

- You can't get what you want

- You get something else instead
- You don't know that you've done it
- You can't fix it after it is done

4 Synthetic Fold and Basin–Range Models

To make aliasing unambiguous, the notebook first uses synthetic geological models.

4.1 Anticline–syncline model

The folded interface is defined as

$$z(x) = z_0 + A \sin\left(\frac{2\pi x}{\lambda}\right), \quad (3)$$

with wavelength $\lambda = 20$ km. A pedagogical forward model is used in which the gravity response is proportional to interface depth:

$$g(x) \propto z(x). \quad (4)$$

This model is not physically exact, but it preserves the correct relationship between geological wavelength and gravity wavelength. It is sufficient for illustrating sampling and aliasing.

4.2 Basin–and–Range style model

A more realistic gravity field is constructed as the superposition of two spatial scales:

$$z(x) = z_0 + A_s \sin\left(\frac{2\pi x}{\lambda_s}\right) + A_d \sin\left(\frac{2\pi x}{\lambda_d}\right), \quad (5)$$

where:

- $\lambda_s \sim 20$ km represents individual basins and ranges,
- $\lambda_d \sim 120$ km represents deep crustal or lithospheric structure.

The resulting gravity profile mimics real Basin-and-Range Bouguer gravity: short-wavelength basin anomalies ride on a strong long-wavelength regional field.

5 Detrending and Its Limitations

Polynomial detrending removes long-wavelength components of a signal. In spectral terms, it acts as a crude high-pass filter. However:

- detrending is not physically motivated,
- high-order detrending can remove real geological signal,
- detrending does not correct aliasing.

The notebook uses detrending only as a stepping stone toward a physically based regional correction.

6 Key Takeaways

- Station spacing sets a hard limit on geological resolution.
- Aliasing produces coherent but incorrect structure.
- Isostatic correction is a physical hypothesis expressed as a Fourier filter.

- Spectral methods do not protect against poor sampling.
- Interpretation begins with survey design, not inversion.